RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Tuesday, 25 August 2026, 9:30 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 9:31 AM
Time taken	1 min 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count = 0;
    int c = 0;

    int i = n / 2;
    count++;                    // i initialization

    while (1) {
        count++;                // i < n
        if (i >= n) break;

        int j = 1;
        while (1) {
            count++;            // j < n
            if (j >= n) break;

            int k = 1;
            while (1) {
                count++;        // k < n
                if (k >= n) break;

                c++;
                count++;        // c++

                k = k * 2;
            }

            j = 2 * j;
        }

        i++;
    }
```

```
39      printf("%d", count);
40      return 0;
41  }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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